9

MHD equilibria

9.1 Why use MHD?

Of the three levels of plasma description – Vlasov, two-fluid, and MHD – Vlasov is the most accurate and MHD is the least accurate. So, why use MHD? The answer is that, because MHD is a more macroscopic point of view, it is more efficient to use MHD in situations where the greater detail and accuracy of the Vlasov or two-fluid models are unnecessary. MHD is particularly suitable for situations having complex geometry because it is very difficult to model such situations using the microscopically oriented Vlasov or two-fluid approaches and because geometrical complexities are often most important at the MHD level of description. The equilibrium and gross stability of three-dimensional, finite-extent plasma configurations are typically analyzed using MHD. Issues requiring a two-fluid or a Vlasov point of view can exist and be important, but these more subtle questions can be addressed after an approximate understanding has first been achieved using MHD. The MHD point of view is especially relevant to situations where magnetic forces are used to confine or accelerate plasmas or liquid conductors such as molten metals. Examples of such situations include magnetic fusion confinement plasmas, solar and astrophysical plasmas, planetary and stellar dynamos, arcs, and magnetoplasmadynamic thrusters. Although molten metals are not plasmas, they are described by MHD and, in fact, the MHD description is actually more appropriate and more accurate for molten metals than it is for plasmas.

We begin our discussion by examining certain general properties of magnetic fields in order to develop an intuitive understanding of the various stresses governing MHD equilibrium and stability. The MHD equation of motion,

$$\rho\left[\frac{\partial \mathbf{U}}{\partial t}+\mathbf{U}\cdot\nabla\mathbf{U}\right]=\mathbf{J}\times\mathbf{B}-\nabla P, \tag{9.1}$$

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is a generalization of the equation of motion for an ordinary fluid because it includes the $\mathbf{J} \times \mathbf{B}$ magnetic force. Plasma viscosity is normally very small and is usually omitted from the MHD equation of motion. However, when torques exist, a viscous damping term needs to be included if one wishes to consider equilibria. This is because viscous damping is required to balance any torque in equilibrium; otherwise the plasma will spin up without limit. Such situations will be discussed in Section 9.9; until then, viscosity will be assumed to be negligible and will be omitted from Eq. (9.1).

9.2 Vacuum magnetic fields

The simplest non-trivial magnetic field results when the entire magnetic field is produced by electric currents located outside the volume of interest so there are no currents in the volume of interest. This type of magnetic field is called a vacuum field since it could exist in a vacuum. Because there are no local currents, a vacuum field satisfies

$$\nabla \times \mathbf{B}_{vac} = 0. \tag{9.2}$$

Since the curl of a gradient is always zero, a vacuum field must be the gradient of some scalar potential χ, i.e., the vacuum field can always be expressed as $\mathbf{B}_{vac} = \nabla\chi$. For this reason vacuum magnetic fields are also called potential magnetic fields. Because all magnetic fields must satisfy $\nabla \cdot \mathbf{B} = 0$, the potential χ satisfies Laplace's equation,

$$\nabla^2\chi = 0. \tag{9.3}$$

Hence the entire mathematical theory of vacuum electrostatic fields can be brought into play when studying vacuum magnetic fields. Vacuum electrostatic theory shows that if either χ or its normal derivative is specified on the surface S bounding a volume V, then χ is uniquely determined in V. Also, if an equilibrium configuration has symmetry in some direction so that the coefficients of the relevant linearized partial differential equations do not depend on this direction, the linearized equations may be Fourier transformed in this "ignorable" direction. Vacuum is automatically symmetric in all directions and Poisson's equation reduces to Laplace's equation, which is intrinsically linear. The linearity of the equation and the symmetry of the physical medium cause Laplace's equation to reduce to one of the standard equations of mathematical physics. For example, consider a cylindrical configuration with coordinates r, θ, and z and suppose this configuration is axially and azimuthally uniform so that both θ and z are ignorable coordinates; i.e., the coefficients of the partial differential equation do not depend on θ or on z. Fourier analysis of Eq. (9.3) implies χ can be expressed as the linear

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superposition of modes varying as $\exp(\mathrm{i}m\theta+\mathrm{i}kz)$. For each choice of m and k, Eq. (9.3) becomes

$$\frac{\partial^2\chi}{\partial r^2}+\frac{1}{r}\frac{\partial\chi}{\partial r}-\left(\frac{m^2}{r^2}+k^2\right)\chi=0. \tag{9.4}$$

Defining $s=kr$, this can be recast as

$$\frac{\partial^2\chi}{\partial s^2}+\frac{1}{s}\frac{\partial\chi}{\partial s}-\left(1+\frac{m^2}{s^2}\right)\chi=0, \tag{9.5}$$

a modified Bessel's equation. The solutions of Eq. (9.5) are the modified Bessel functions $I_m(kr)$ and $K_m(kr)$. Thus, the general solution of Laplace's equation here is

$$\chi(r,\theta,z)=\sum_{m=-\infty}^{\infty}\int\left[a_m(k)I_m(kr)+b_m(k)K_m(kr)\right]\mathrm{e}^{\mathrm{i}m\theta+\mathrm{i}kz}\mathrm{d}k, \tag{9.6}$$

where the coefficients $a_m(k)$, $b_m(k)$ are determined by specifying either χ or its normal derivative on the bounding surface. Analogous solutions can be found in geometries having other symmetries.

This behavior can be viewed in a more general way. Equation (9.3) states that the sum of partial second derivatives in two or three different directions is zero, so at least one of these terms must be negative and at least one term must be positive. Since negative χ''/χ corresponds to oscillatory (harmonic) behavior and positive χ''/χ corresponds to exponential (non-harmonic) behavior, any solution of Laplace's equation must be oscillatory in one or two directions (the θ and z directions for the cylindrical example here), and exponentially growing or decaying in the remaining direction or directions (the r direction for the cylindrical example here).

Non-vacuum magnetic fields are more complicated than vacuum fields and, unlike vacuum fields, are not uniquely determined by the surface boundary conditions. This is because non-vacuum fields are determined by both the current distribution within the volume and the surface boundary conditions. Vacuum fields are distinguished from non-vacuum fields because vacuum fields are the lowest energy fields satisfying given boundary conditions on the surface S of a volume V. Let us now prove this statement.

Consider a volume V bounded by a surface S over which boundary conditions are specified. Let $\mathbf{B}_{\min}(\mathbf{r})$ be the magnetic field having the *lowest* stored magnetic energy of all possible magnetic fields satisfying the prescribed boundary conditions. We use methods of variational calculus to prove this lowest energy field is the vacuum field.

Consider some slightly different field, denoted as $\mathbf{B}(\mathbf{r})$, that satisfies the same boundary conditions as $\mathbf{B}_{\min}$. This slightly different field can be expressed as

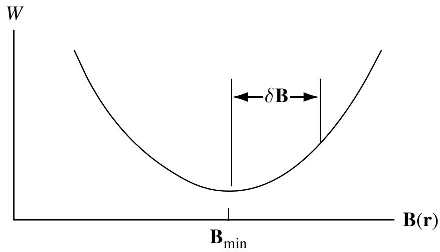

Fig. 9.1 Magnetic field energy for different configurations. $\mathbf{B}_{\min}$ is the configuration with minimum magnetic energy for a given boundary condition.

$\mathbf{B}(\mathbf{r}) = \mathbf{B}_{\min}(\mathbf{r}) + \delta\mathbf{B}(\mathbf{r})$, where $\delta\mathbf{B}(\mathbf{r})$ is a small arbitrary variation about $\mathbf{B}_{\min}$. This situation is sketched in a qualitative fashion in Fig. 9.1, where the horizontal axis represents a continuum of different allowed choices for the vector function $\mathbf{B}(\mathbf{r})$. Since $\mathbf{B}(\mathbf{r})$ satisfies the same boundary conditions as $\mathbf{B}_{\min}$, $\delta\mathbf{B}(\mathbf{r})$ must vanish on S. The magnetic energy W associated with $\mathbf{B}(\mathbf{r})$ can be evaluated as

$$\begin{aligned} 2\mu_0 W &= \int_V (\mathbf{B}_{\min} + \delta\mathbf{B})^2 \, \mathrm{d}^3 r \\ &= \int_V \mathbf{B}_{\min}^2 \mathrm{d}^3 r + 2\int_V \delta\mathbf{B}\cdot\mathbf{B}_{\min} \mathrm{d}^3 r + \int_V (\delta\mathbf{B})^2 \, \mathrm{d}^3 r. \end{aligned} \tag{9.7}$$

If the middle term does not vanish, $\delta\mathbf{B}$ could always be chosen to be antiparallel to $\mathbf{B}_{\min}$ inside V, in which case $\delta\mathbf{B}\cdot\mathbf{B}_{\min}$ would be negative. Since $\delta\mathbf{B}$ is assumed small, $\delta\mathbf{B}\cdot\mathbf{B}_{\min}$ would be larger in magnitude than $(\delta\mathbf{B})^2$. Such a choice for $\delta\mathbf{B}$ would make W lower than the energy of the assumed lowest energy field, thus contradicting the assumption that $\mathbf{B}_{\min}$ is the lowest energy field. Thus, the only way to ensure that $\mathbf{B}_{\min}$ is indeed the true minimum is to require

$$\int_v \delta\mathbf{B}\cdot\mathbf{B}_{\min} \mathrm{d}^3 r = 0 \tag{9.8}$$

no matter how $\delta\mathbf{B}$ is chosen. Using $\delta\mathbf{B} = \nabla\times\delta\mathbf{A}$ and the vector identity $\nabla\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot\nabla\times\mathbf{A} - \mathbf{A}\cdot\nabla\times\mathbf{B}$, Eq. (9.8) can be integrated by parts to obtain

$$\int_V [\nabla\cdot(\delta\mathbf{A}\times\mathbf{B}_{\min}) + \delta\mathbf{A}\cdot\nabla\times\mathbf{B}_{\min}] \, \mathrm{d}^3 r = 0. \tag{9.9}$$

The first term can be transformed into a surface integral over S using Gauss' theorem. This surface integral vanishes because $\delta\mathbf{A}$ must vanish on the bounding surface (recall that the variation satisfies the same boundary condition as the minimum energy field). Because $\delta\mathbf{B}$ is arbitrary within V, $\delta\mathbf{A}$ must also be arbitrary within V and so the only way for the second term in Eq. (9.9) to vanish is to have $\nabla\times\mathbf{B}_{\min} = 0$. Thus, $\mathbf{B}_{\min}$ must be a vacuum field.

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An important corollary is as follows: suppose boundary conditions are specified on the surface enclosing some volume. These boundary conditions can be considered as "rules" that must be satisfied by any solution to the equations. All configurations satisfying the imposed boundary condition and having finite current within the volume are *not* in the lowest energy state. Thus, non-vacuum fields can, in principle, have free energy available for driving boundary-condition-preserving instabilities.

9.3 Force-free fields

Although the vacuum field is the lowest energy configuration satisfying prescribed boundary conditions, non-vacuum configurations do not always "decay" to this lowest energy state. This is because there also exists a family of higher energy configurations to which the system may decay; these are the so-called force-free states. The current is not zero in a force-free state but the magnetic force is zero because the current density is everywhere parallel to the magnetic field. Thus $\mathbf{J} \times \mathbf{B}$ vanishes even though both $\mathbf{J}$ and $\mathbf{B}$ are finite. If a plasma not initially in a force-free state somehow evolves towards a force-free state, it will become "stuck" in the force-free state because, by definition, no forces can act on the plasma to move it out of the force-free state. The magnetic energy of a force-free field is not the absolute minimum energy for the specified boundary conditions, but it is a local minimum in configuration space as sketched in Fig. 9.2. This hierarchy of states is somewhat analogous to the states of a quantum system – the vacuum field is the analog of the ground state and the force-free states are the analogs of higher energy quantum states.

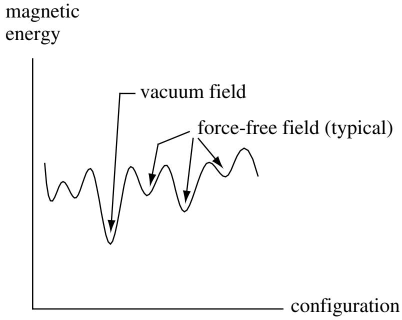

Fig. 9.2 Sketch of magnetic energy dependence on configuration of a system with fixed, specified boundary conditions. The variation of the configuration would correspond to different internal current profiles. The force-free configurations are local energy minima while the vacuum configuration has the absolute lowest minimum.

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9.4 Magnetic pressure and tension

Much useful insight can be obtained by considering the force between two parallel current-carrying wires as shown in Fig. 9.3. Calculation of the field **B** observed at one wire due to the current in the other shows that the $\mathbf{J} \times \mathbf{B}$ force is such as to push the wires together; i.e., parallel currents attract each other. Conversely, antiparallel currents repel each other.

A bundle of parallel wires as shown in Fig. 9.4 will therefore mutually attract each other resulting in an effective net force that acts to reduce the diameter of the bundle. The bundle could be replaced by a distributed current such as the current carried by a finite-radius, cylindrical plasma. This contracting, inward-directed force is called the pinch force or the pinch effect.

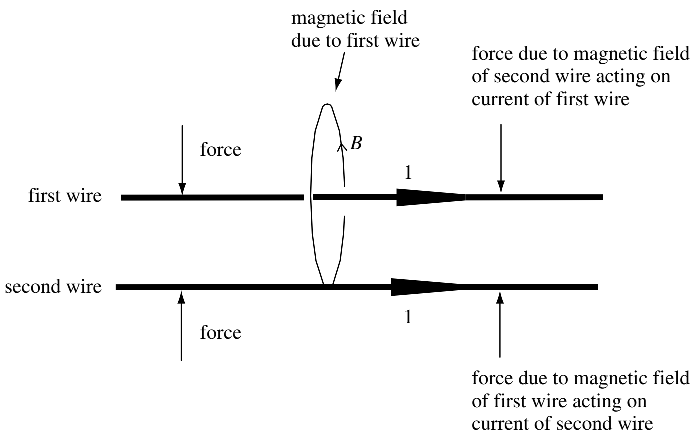

Fig. 9.3 Parallel currents attract each other.

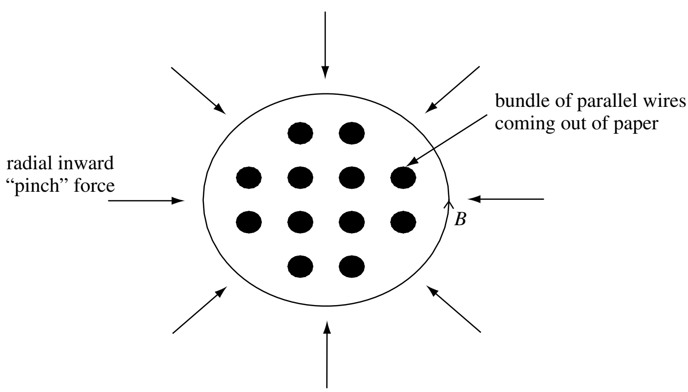

Fig. 9.4 Bundle of currents attract each other giving effective radial inward pinch force.

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The pinch force may be imagined as being due to a "tension" in the azimuthal magnetic field that wraps around the distributed current. Extending this metaphor, the azimuthal magnetic field is visualized as acting like an "elastic band" which encircles the distributed current and squeezes or pinches the current to a smaller diameter. This concept is consistent with the situation of two permanent magnets attracting each other when the respective north and south poles face each other. The magnetic field lines go from the north pole of one magnet to the south pole of the other and so one can imagine that the attraction of the two magnets is due to tension in the field lines spanning the gap between the two magnets.

Now consider a current-carrying loop such as is shown in Fig. 9.5. Because the currents on opposite sides of the loop flow in opposing directions, there will be a repulsive force between the current element at each point and the current element on the opposite side of the loop. The net result is a force directed to expand the diameter of the loop. This force is called the hoop force or hoop stress. Hoop force may also be interpreted metaphorically by introducing the concept of magnetic pressure. As shown in Fig. 9.5, the magnetic field lines linking the current are more dense inside the loop than outside, a purely geometrical effect resulting from the curvature of the current path. Since magnetic field strength is proportional to field line density, the magnetic field is stronger on the inside of the loop than on the outside, i.e., B^2 is stronger on the inside than on the outside. The magnetic field in the plane of the loop is normal to the plane and so one can explain the hoop force by assigning a magnetic "pressure" proportional to B^2 acting in the direction perpendicular to $\mathbf{B}$. Because B^2 is stronger on the inside of the loop than on the outside, there is a larger magnetic pressure on the inside. The outward force due to this pressure imbalance is consistent with the hoop force.

The combined effects of magnetic pressure and tension can be visualized by imagining a current-carrying loop where the conductor has a finite diameter. Suppose this loop initially has the relative proportions of an automobile tire, as shown in Fig. 9.6. The hoop force will cause the diameter of the loop to increase, while the pinch force will cause the diameter of the conductor to decrease. The net

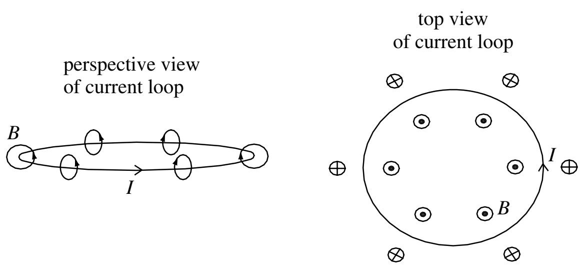

Fig. 9.5 Perspective and top view of magnetic fields generated by a current loop.

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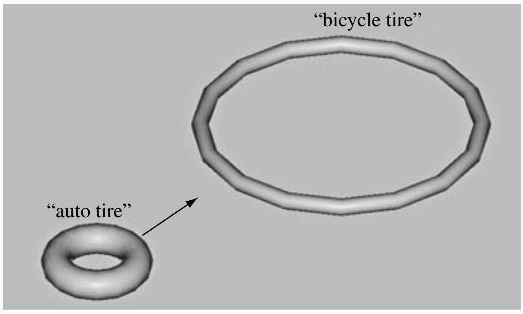

Fig. 9.6 Magnetic forces act to transform a fat, small current loop (auto tire) into a large, skinny loop (bicycle tire).

result is that these combined forces will cause the automobile tire to evolve towards the relative proportions of a bicycle tire (large major radius, small minor radius).

9.5 Magnetic stress tensor

The existence of magnetic pressure and tension shows that the magnetic force is different in different directions, and so the magnetic force ought to be characterized by an anisotropic stress tensor. To establish this mathematically, the vector identity $\nabla B^2/2 = \mathbf{B}\cdot\nabla\mathbf{B} + \mathbf{B}\times\nabla\times\mathbf{B}$ is invoked so that the magnetic force can be expressed as

$$\begin{aligned}\mathbf{J}\times\mathbf{B} &= \frac{1}{\mu_0}(\nabla\times\mathbf{B})\times\mathbf{B} \\ &= \frac{1}{\mu_0}\left[-\nabla\left(\frac{B^2}{2}\right)+\mathbf{B}\cdot\nabla\mathbf{B}\right] \\ &= -\frac{1}{\mu_0}\nabla\cdot\left[\frac{B^2}{2}\mathbf{I}-\mathbf{BB}\right],\end{aligned} \tag{9.10}$$

where $\mathbf{I}$ is the unit tensor and the relation $\nabla\cdot(\mathbf{BB}) = (\nabla\cdot\mathbf{B})\,\mathbf{B}+\mathbf{B}\cdot\nabla\mathbf{B} = \mathbf{B}\cdot\nabla\mathbf{B}$ has been used. At any point $\mathbf{r}$ a local Cartesian coordinate system can be defined with z axis parallel to the local value of $\mathbf{B}$ so that Eq. (9.1) can be written as

$$\rho\left[\frac{\partial\mathbf{U}}{\partial t}+\mathbf{U}\cdot\nabla\mathbf{U}\right] = -\nabla\cdot\begin{bmatrix} P+\dfrac{B^2}{2\mu_0} & & \\ & P+\dfrac{B^2}{2\mu_0} & \\ & & P-\dfrac{B^2}{2\mu_0}\end{bmatrix} \tag{9.11}$$

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showing again that the magnetic field acts like a pressure in the directions transverse to **B** (i.e., x, y directions in the local Cartesian system) and like a tension in the direction parallel to **B**.

While the above interpretation is certainly useful, it can be somewhat misleading because it might be interpreted as implying the existence of a force in the direction of **B** when in fact no such force exists because $\mathbf{J}\times\mathbf{B}$ clearly does not have a component in the **B** direction. A more accurate way to visualize the relation between magnetic pressure and tension is to rearrange the second line of Eq. (9.10) as

$$\mathbf{J}\times\mathbf{B} = \frac{1}{\mu_0}\left[-\nabla\left(\frac{B^2}{2}\right) + B^2\hat{B}\cdot\nabla\hat{B} + \hat{B}\hat{B}\cdot\nabla\left(\frac{B^2}{2}\right)\right] = \frac{1}{\mu_0}\left[-\nabla_\perp\left(\frac{B^2}{2}\right) + B^2\boldsymbol{\kappa}\right] \tag{9.12}$$

or

$$\mathbf{J}\times\mathbf{B} = \frac{1}{\mu_0}\left[-\nabla_\perp\left(\frac{B^2}{2}\right) + B^2\boldsymbol{\kappa}\right]. \tag{9.13}$$

Here

$$\boldsymbol{\kappa} = \hat{B}\cdot\nabla\hat{B} = -\frac{\hat{R}}{R} \tag{9.14}$$

is a measure of the curvature of the magnetic field at a selected point on a field line and, in particular, **R** is the local radius of curvature vector. The vector **R** goes from the center of curvature to the selected point on the field line. The $\boldsymbol{\kappa}$ term in Eq. (9.13) describes a force that tends to straighten out magnetic curvature and is a more precise way for characterizing field line tension (recall that tension similarly acts to straighten out curvature). The term involving $\nabla_\perp B^2$ portrays a magnetic force due to pressure gradients perpendicular to the magnetic field and is a more precise expression of the hoop force.

In our earlier discussion it was shown that the vacuum magnetic field is the lowest energy state of all fields satisfying prescribed boundary conditions. A vacuum field might be curved for certain boundary conditions (e.g., a permanent magnet with finite dimensions) and, in such a case, the two terms in Eq. (9.13) are both finite but exactly cancel each other. We can think of the minimum-energy state for given boundary conditions as being analogous to the equilibrium state of a system of stiff rubber hoses that have been pre-formed into shapes having the morphology of the vacuum magnetic field and that have their ends fixed at the bounding surface. Currents will cause the morphology of this system to deviate from the equilibrium state. However, since any deformation requires work to be done on the system, currents invariably cause the system to be in a higher energy state.

9.6 Flux preservation, energy minimization, and inductance

Another useful way of understanding magnetic field behavior relates to the concept of electric circuit inductance. The self-inductance L of a circuit component is defined as the magnetic flux Φ linking the component divided by the current I flowing through the component, i.e.,

$$L = \frac{\Phi}{I}. \tag{9.15}$$

Consider an arbitrary short-circuited coil with current I located in an infinite volume V and let C denote the three-dimensional spatial contour traced out by the wire constituting the coil. The total magnetic flux linked by the turns of the coil can be expressed as

$$\Phi = \int \mathbf{B} \cdot \mathrm{d}\mathbf{s} = \int \nabla \times \mathbf{A} \cdot \mathrm{d}\mathbf{s} = \oint_C \mathbf{A} \cdot \mathrm{d}\mathbf{l}, \tag{9.16}$$

where the surface integral is over the area elements linked by the coil turns. The energy contained in the magnetic field produced by the coil is

$$\begin{aligned} W &= \int_V \frac{B^2}{2\mu_0} \mathrm{d}^3 r \\ &= \frac{1}{2\mu_0} \int_V \mathbf{B} \cdot \nabla \times \mathbf{A} \mathrm{d}^3 r. \end{aligned} \tag{9.17}$$

However, using the vector identity $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot \nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mathbf{B}$ this magnetic energy can be expressed as

$$W = \frac{1}{2\mu_0} \int_V \mathbf{A} \cdot \nabla \times \mathbf{B} \mathrm{d}^3 r + \frac{1}{2\mu_0} \int_{S_\infty} \mathrm{d}\mathbf{s} \cdot \mathbf{A} \times \mathbf{B}, \tag{9.18}$$

where Gauss' law has been invoked to obtain the second term, an integral over the surface at infinity, S_∞. This surface integral vanishes because (i) at infinity the magnetic field must fall off at least as fast as a dipole, i.e., $B \sim R^{-3}$, where R is the distance to the origin, (ii) the vector potential magnitude A scales as the integral of B so $A \sim R^{-2}$, and (iii) the surface at infinity scales as R^2.

Using Ampère's law, Eq. (9.18) can thus be rewritten as

$$W = \frac{1}{2} \int_V \mathbf{A} \cdot \mathbf{J} \mathrm{d}^3 r. \tag{9.19}$$

However, $\mathbf{J}$ is only finite in the coil wire and so the integral reduces to an integral over the volume of the wire. A volume element of wire can be expressed as

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$d^3r = d\mathbf{s} \cdot d\mathbf{l}$, where $d\mathbf{l}$ is an element of length along the wire, and $d\mathbf{s}$ is the cross-sectional area of the wire. Since $\mathbf{J}$ and $d\mathbf{l}$ are parallel, they can be interchanged in Eq. (9.19) which becomes

$$\begin{aligned} W &= \frac{1}{2}\int_{V_{coil}} \mathbf{A} \cdot d\mathbf{l}\ \mathbf{J} \cdot d\mathbf{s} \\ &= \frac{I}{2}\oint \mathbf{A} \cdot d\mathbf{l} \\ &= \frac{I\Phi}{2} \\ &= \frac{\Phi^2}{2L} \\ &= \frac{1}{2}LI^2, \end{aligned} \tag{9.20}$$

where $\mathbf{J} \cdot d\mathbf{s} = I$ is the current flowing through the wire. Thus, the energy stored in the magnetic field produced by a coil is just the inductive energy of the coil.

If the coil is perfectly short-circuited, then it must be flux conserving, for if there were a change in flux, a voltage would appear across the ends of the coil. A closed current flowing in a perfectly conducting plasma is thus equivalent to a short-circuited current-carrying coil and so the perfectly conducting plasma can be considered as a flux-conserver. If flux is conserved, i.e., $\Phi = const.$, the second from last line in Eq. (9.20) shows that *the magnetic energy of the system will be lowered by any rearrangement of circuit topology that increases self-inductance*.

Hot plasmas are reasonably good flux conservers because of their high electrical conductivity. Thus, any inductance-increasing change in the topology of plasma currents will release free energy, which could be used to drive an instability. Since forces act so as to reduce the potential energy of a system, magnetic forces due to current flowing in a plasma will always act so as to increase the self-inductance of the configuration. One can therefore write the force $\mathbf{F}$ due to a flux-conserving change in inductance L as

$$\mathbf{F} = -\frac{\Phi^2}{2}\nabla\left(\frac{1}{L}\right). \tag{9.21}$$

The pinch force is consistent with this interpretation since the inductance of a conductor depends inversely on its radius. The hoop force is also consistent with this interpretation since inductance of a current loop increases with the major radius of the loop. More complicated behavior can also be explained, especially the kink instability to be discussed later. In the kink instability, current initially flowing in a straight line develops an instability that causes the current path to become helical. Since a coil (helix) has more inductance than a straight length of the same wire, the effect of the kink instability also acts so as to increase the circuit self-inductance.

9.7 Static versus dynamic equilibria

We define (i) a static equilibrium to be a time-independent solution to Eq. (9.1) having no flow velocity and (ii) a dynamic equilibrium as a solution with steady-state flow velocities. Thus, $\mathbf{U} = 0$ for a static equilibrium whereas $\mathbf{U}$ is finite and steady-state for a dynamic equilibrium. For a static equilibrium the MHD equation of motion reduces to

$$\nabla P = \mathbf{J} \times \mathbf{B}. \tag{9.22}$$

For a dynamic equilibrium the MHD equation of motion reduces to

$$\rho \mathbf{U} \cdot \nabla \mathbf{U} + \nabla P = \mathbf{J} \times \mathbf{B} + \nu \rho \nabla^2 \mathbf{U}, \tag{9.23}$$

where the last term represents a viscous damping and ν is the kinematic viscosity. By taking the curl of these last two equations we see that for static equilibria the magnetic force must be conservative, i.e., $\nabla \times (\mathbf{J} \times \mathbf{B}) = 0$, whereas for dynamic equilibria the magnetic force is typically not conservative since in general $\nabla \times (\mathbf{J} \times \mathbf{B}) \neq 0$. Thus, the character of the magnetic field is quite different for the two cases. Static equilibria are relevant to plasma confinement devices such as tokamaks, stellarators, reversed field pinches, and spheromaks, while dynamic equilibria are mainly relevant to arcs, jets, and magnetoplasmadynamic thrusters, but can also be relevant to tokamaks, etc., if there are flows. Both static and dynamic equilibria occur in space plasmas.

9.8 Static equilibria

9.8.1 Static equilibria in two dimensions: the Bennett pinch

The simplest static equilibrium was first investigated by Bennett (1934) and is called the Bennett pinch or z-pinch (here z refers to the direction of the current). This configuration, sketched in Fig. 9.7, consists of an infinitely long axisymmetric cylindrical plasma with axial current density $J_z = J_z(r)$ and no other currents.

The axial current flowing within a circle of radius r is

$$I(r) = \int_0^r 2\pi r' J_z(r') \mathrm{d}r' \tag{9.24}$$

and the axial current density is related to this integrated current by

$$J_z(r) = \frac{1}{2\pi r} \frac{\partial I}{\partial r}. \tag{9.25}$$

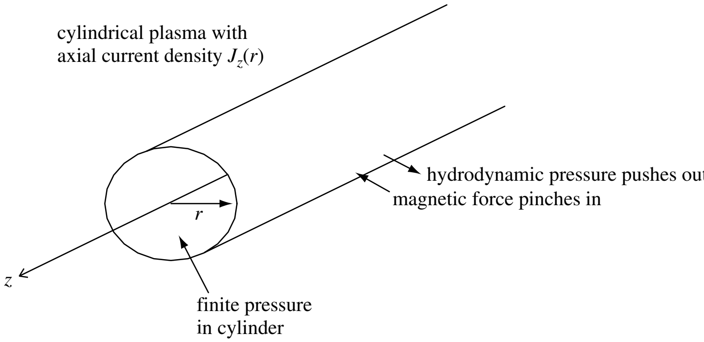

Fig. 9.7 Geometry of Bennett pinch.

Since Ampère's law gives

$$B_\theta(r) = \frac{\mu_0 I(r)}{2\pi r}, \tag{9.26}$$

Eq. (9.22) can be written

$$r^2 \frac{\partial P}{\partial r} = -\frac{\mu_0}{8\pi^2}\frac{\partial I^2}{\partial r}. \tag{9.27}$$

Integrating Eq. (9.27) from $r = 0$ to $r = a$, where a is the outer radius of the cylindrical plasma, gives the relation

$$\int_0^a r^2 \frac{\partial P}{\partial r} \mathrm{d}r = \left[r^2 P(r)\right]_0^a - 2\int_0^a rP(r)\mathrm{d}r = -\frac{\mu_0 I^2(a)}{8\pi^2}. \tag{9.28}$$

The integrated term vanishes at both $r = 0$ and $r = a$ since $P(a) = 0$ by definition. If the temperature is uniform, the pressure can be expressed as $P(r) = n(r)\kappa T$ and so Eq. (9.28) can be expressed as

$$I^2 = \frac{8\pi N\kappa T}{\mu_0}, \tag{9.29}$$

where $N = \int_0^a n(r)2\pi r\mathrm{d}r$ is the number of particles per axial length. Equation (9.29), called the Bennett relation, shows that the current required to confine a given N and T is independent of the details of the internal density profile. This relation describes the simplest non-trivial MHD equilibrium and suggests that quite modest currents could contain substantial plasma pressures. This relation motivated the design of early magnetic fusion confinement devices but, as will be seen, it is overly optimistic because simple z-pinch equilibria turn out to be highly unstable.

Confinement using currents flowing in the azimuthal direction is also possible, but this configuration, known as a θ-pinch, is fundamentally transient. In a θ-pinch, a rapidly changing azimuthal current in a coil surrounding a cylindrical

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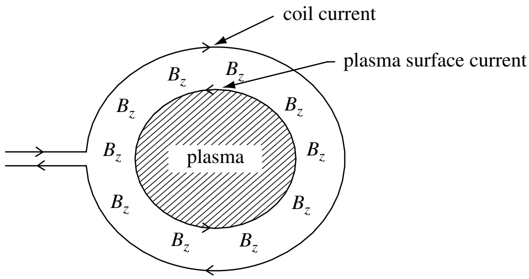

Fig. 9.8 Theta pinch configuration. Rapidly changing azimuthal coil current induces equal and opposite azimuthal current in plasma surface. Pressure of B_z between these two currents pushes in on plasma and provides confinement.

plasma creates a transient B_z field as shown in Fig. 9.8. Because the conducting plasma conserves magnetic flux, the transient B_z field cannot penetrate the plasma and so is confined to the vacuum region between the plasma and the coil. This exclusion of the B_z field by the plasma requires the existence in the plasma surface of an induced azimuthal current that creates in the plasma interior a magnetic field, which exactly cancels the coil-produced transient B_z field. The radial confining force results from a radially inward force $\mathbf{J} \times \mathbf{B} = J_\theta \hat{\theta} \times B_z \hat{z}$, where B_z is the magnetic field associated with the current in the coil and J_θ is the plasma surface current. An alternative but equivalent point of view is to invoke the concept that opposite currents repel and argue that the azimuthal current in the coil repels the oppositely directed azimuthal current in the plasma surface, thereby pushing the plasma inwards and so balancing the outwards force due to plasma pressure. The θ-pinch configuration is necessarily transient, because the induced azimuthally directed surface current cannot be sustained in steady state.

9.8.2 Impossibility of self-confinement of current-carrying plasma in three dimensions: the virial theorem

The Bennett analysis showed that axial currents flowing in an infinitely long cylindrical plasma generate a pinch force, which confines a finite pressure plasma. The inward pinch force balances the outward force associated with the pressure gradient. The question now is whether this two-dimensional result can be extended to three dimensions; i.e., is it possible to have a finite-pressure three-dimensional plasma, as shown in Fig. 9.9, that is confined entirely by currents circulating within the plasma? To be more specific, is it possible to have a finite-radius plasma sphere surrounded by vacuum where the confinement of the finite plasma pressure is entirely provided by the magnetic force of currents circulating in the plasma; i.e., can the plasma hold itself together by its own "bootstraps?" The

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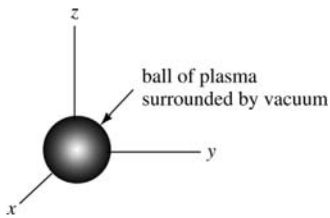

Fig. 9.9 Plasma sphere with finite pressure surrounded by vacuum.

answer, a resounding "no," is provided by a virial theorem due to Shafranov (1966). A virial is a suitably weighted integral over the entirety of a system and contains information about how an extensive property such as energy is partitioned in the system. For example, in mechanics the virial can be the time average of the potential or of the kinetic energy.

The MHD virial theorem is obtained by supposing a self-confining configuration exists and then showing this supposition leads to a contradiction.

We therefore postulate the existence of a spherical plasma with the following properties:

1. the plasma has finite radius a and is surrounded by vacuum,
2. the plasma is in static MHD equilibrium,
3. the plasma has finite internal pressure and the pressure gradient is entirely balanced by magnetic forces due to currents circulating in the plasma; i.e., there are no currents in the surrounding vacuum region.

Using Eqs. (9.10) and (9.11) the static MHD equilibrium can be expressed as

$$\nabla \cdot \mathbf{T} = 0, \tag{9.30}$$

where the tensor $\mathbf{T}$ is defined as

$$\mathbf{T} = \left(P + \frac{B^2}{2\mu_0}\right)\mathbf{I} - \frac{1}{\mu_0}\mathbf{BB}. \tag{9.31}$$

Let $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$ be the vector from the center of the plasma to the point of observation and consider the virial expression

$$\begin{aligned} \nabla \cdot (\mathbf{T} \cdot \mathbf{r}) &= \sum_{jk} \frac{\partial}{\partial x_j}\left(T_{jk} x_k\right) \\ &= (\nabla \cdot \mathbf{T}) \cdot \mathbf{r} + \sum_{jk} T_{jk} \frac{\partial}{\partial x_j} x_k \\ &= \text{Trace } \mathbf{T}, \end{aligned} \tag{9.32}$$

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where Trace $\mathbf{T} = \sum_{jk} T_{jk}\delta_{jk}$. From Eq. (9.31) (or, equivalently from the matrix form in Eq. (9.11)) it is seen that

$$\text{Trace } \mathbf{T} = 3P + B^2/2\mu_0$$

is *positive-definite*.

We now integrate both sides of Eq. (9.32) over all space. Since the right-hand side of Eq. (9.32) is positive definite, the integral of the right-hand side over all space is finite and positive. The integral of the left-hand side can be transformed to a surface integral at infinity using Gauss' theorem

$$\begin{aligned}\int \mathrm{d}^3 r \nabla\cdot(\mathbf{T}\cdot\mathbf{r}) &= \int_{S_\infty} \mathrm{d}\mathbf{s}\cdot\mathbf{T}\cdot\mathbf{r} \\ &= \int_{S_\infty} \mathrm{d}\mathbf{s}\cdot\left[\left(P + \frac{B^2}{2\mu_0}\right)\mathbf{I} - \frac{1}{\mu_0}\mathbf{BB}\right]\cdot\mathbf{r}.\end{aligned} \tag{9.33}$$

Since the plasma is assumed to have finite extent, P is zero on the surface at infinity. The magnetic field can be expanded in multipoles with the lowest order multipole being a dipole. The magnetic field of a dipole scales as r^{-3} for large r while the surface area $\int \mathrm{d}s$ scales as r^2. Thus the left-hand side term scales as $\int \mathrm{d}sB^2 r \sim r^{-3}$ and so vanishes as $r \to \infty$. This is in contradiction to the right-hand side being positive definite and so the set of initial assumptions must be erroneous. Thus, any finite extent, three-dimensional, static plasma equilibrium must involve at least some currents external to the plasma. A finite extent, three-dimensional static plasma can therefore only be in equilibrium if at least some of the magnetic field is produced by currents in coils that are both external to the plasma and held in place by some mechanical structure. If the coils were not supported by a mechanical structure, then the current in the coils could be considered as part of the MHD plasma and the virial theorem would be violated. In summary, a finite extent three-dimensional plasma in static equilibrium with a finite internal hydrodynamic pressure P must ultimately have some tangible exterior object to "push against." The buttressing is provided by magnetic forces acting between currents in external coils and currents in the plasma.

9.8.3 Three-dimensional static equilibria: the Grad–Shafranov equation

Despite the simple appearance of Eq. (9.22), its three-dimensional solution is far from trivial. Before even attempting to find a solution, it is important to decide the appropriate way to pose the problem, i.e., it must be decided which quantities are prescribed and which are to be solved for. For example, one might imagine prescribing a pressure profile $P(\mathbf{r})$ and then using Eq. (9.22) to determine a corresponding $\mathbf{B}(\mathbf{r})$ with associated current $\mathbf{J}(\mathbf{r})$; alternatively one could imagine

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prescribing $\mathbf{B}(\mathbf{r})$ and then using Eq. (9.22) to determine $P(\mathbf{r})$. Unfortunately, neither of these approaches work in general because solutions to Eq. (9.22) exist for only a very limited set of functions.

The reason why an arbitrary magnetic field cannot be specified is that $\nabla \times \nabla P = 0$ is always true by virtue of a mathematical identity whereas $\nabla \times (\mathbf{J} \times \mathbf{B}) = 0$ is true only for certain types of $\mathbf{B}(\mathbf{r})$. It is also true that an arbitrary equilibrium pressure profile $P(\mathbf{r})$ cannot be prescribed (for example, see Assignment 2 where it is demonstrated that no $\mathbf{J} \times \mathbf{B}$ force exists that can confine a plasma with a spherically symmetric pressure profile). Equilibria thus exist only for certain specific situations, and these situations typically require symmetry in some direction. We shall now examine a very important example, namely static equilibria that are azimuthally symmetric about an axis (typically defined as the z axis). This symmetry applies to a wide variety of magnetic confinement devices used in magnetic fusion research, for example tokamaks, reversed field pinches, spheromaks, and field reversed theta pinches.

We start the analysis by assuming azimuthal symmetry about the z axis of a cylindrical coordinate system $\{r, \phi, z\}$ so any *physical* quantity f has the property $\partial f / \partial \phi = 0$. Because of the identity $\nabla \times \nabla \phi = 0$, algebraic manipulations become considerably simplified if vectors in the ϕ direction are expressed in terms of $\nabla \phi = \hat{\phi}/r$ rather than in terms of $\hat{\phi}$. As sketched in Fig. 9.10 the term toroidal denotes vectors in the ϕ direction (long way around a torus) and the term poloidal denotes vectors in the $r - z$ plane.

The most general form for an axisymmetric magnetic field is

$$\mathbf{B} = \frac{1}{2\pi} \left(\nabla \psi \times \nabla \phi + \mu_0 I \nabla \phi \right). \tag{9.34}$$

$\psi(r, z)$ is called the poloidal flux and $I(r, z)$ is the current linked by a circle of radius r with center on the axis at axial location z. The toroidal magnetic field then is

$$\mathbf{B}_{tor} = B_\phi \hat{\phi} = \frac{\mu_0 I}{2\pi} \nabla \phi = \frac{\mu_0 I}{2\pi r} \hat{\phi} \tag{9.35}$$

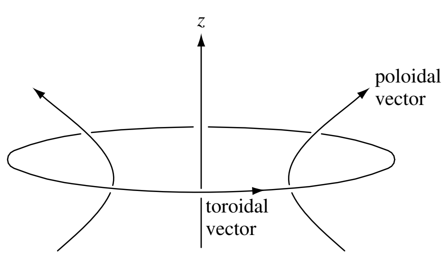

Fig. 9.10 Toroidal vectors and poloidal vectors.

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showing that the functional form of Eq. (9.34) is consistent with Ampère's law $\oint \mathbf{B}\cdot \mathrm{d}\mathbf{l} = \mu_0 I$. The poloidal magnetic field is

$$\mathbf{B}_{pol} = \frac{1}{2\pi}\left(\nabla\psi \times \nabla\phi\right). \tag{9.36}$$

Integration of the poloidal magnetic field over the area of a circle of radius r with center at axial location z gives

$$\int_0^r \mathbf{B}_{pol}\cdot \mathrm{d}\mathbf{s} = \int_0^r \frac{1}{2\pi}\nabla\psi \times \nabla\phi \cdot \hat{z} 2\pi r' \mathrm{d}r' = \psi(r,z); \tag{9.37}$$

thus $\psi(r,z)$ is the poloidal flux at location r, z. The concept of poloidal flux depends on the existence of axisymmetry, which makes it always possible to associate any location r, z with a circular area of radius r centered at $z = 0$.

Axisymmetry also provides a useful relationship between toroidal and poloidal vectors. In particular, the curl of a toroidal vector is poloidal since

$$\nabla \times \mathbf{B}_{tor} = \frac{\mu_0}{2\pi}\nabla I \times \nabla\phi \tag{9.38}$$

and similarly the curl of a poloidal vector is toroidal since

$$\nabla \times \mathbf{B}_{pol} = \nabla \times \left(B_r\hat{r} + B_z\hat{z}\right) = \hat{\phi}\left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r}\right). \tag{9.39}$$

The curl of the poloidal magnetic field is a Laplacian-like operator on ψ since

$$\nabla\phi\cdot\nabla\times\mathbf{B}_{pol} = \nabla\cdot\left(\mathbf{B}_{pol}\times\nabla\phi\right) = \nabla\cdot\left(\frac{1}{2\pi}\left[\nabla\psi\times\nabla\phi\right]\times\nabla\phi\right) = -\frac{1}{2\pi}\nabla\cdot\left(\frac{1}{r^2}\nabla\psi\right), \tag{9.40}$$

a relationship established using the vector identity $\nabla\cdot(\mathbf{F}\times\mathbf{G}) = \mathbf{G}\cdot\nabla\times\mathbf{F} - \mathbf{F}\cdot\nabla\times\mathbf{G}$. Because $\nabla\times\mathbf{B}_{pol}$ is purely toroidal and $\hat{\phi} = r\nabla\phi$, one can write

$$\nabla\times\mathbf{B}_{pol} = -\frac{r^2}{2\pi}\nabla\cdot\left(\frac{1}{r^2}\nabla\psi\right)\nabla\phi. \tag{9.41}$$

Ampère's law states $\nabla\times\mathbf{B} = \mu_0\mathbf{J}$. Thus, from Eqs. (9.38) and (9.41) the respective toroidal and poloidal currents are

$$\mathbf{J}_{tor} = -\frac{r^2}{2\pi\mu_0}\nabla\cdot\left(\frac{1}{r^2}\nabla\psi\right)\nabla\phi \tag{9.42}$$

and

$$\mathbf{J}_{pol} = \frac{1}{2\pi}\nabla I\times\nabla\phi. \tag{9.43}$$

We are now in a position to evaluate the magnetic force in Eq. (9.22). After decomposing the magnet field and current into toroidal and poloidal components, Eq. (9.22) becomes

$$\nabla P = \mathbf{J}_{pol} \times \mathbf{B}_{tor} + \mathbf{J}_{tor} \times \mathbf{B}_{pol} + \mathbf{J}_{pol} \times \mathbf{B}_{pol}. \tag{9.44}$$

The $\mathbf{J}_{pol} \times \mathbf{B}_{pol}$ term is in the toroidal direction and is the only toroidally directed term on the right-hand side of the equation. However, $\partial P/\partial\phi = 0$ because all physical quantities are independent of ϕ and so $\mathbf{J}_{pol} \times \mathbf{B}_{pol}$ must vanish. This implies

$$(\nabla I \times \nabla\phi) \times (\nabla\psi \times \nabla\phi) = 0, \tag{9.45}$$

which further implies that ∇I must be parallel to $\nabla\psi$. An arbitrary displacement $\mathrm{d}\mathbf{r}$ results in respective changes in current and poloidal flux $\mathrm{d}I = \mathrm{d}\mathbf{r} \cdot \nabla I$ and $\mathrm{d}\psi = \mathrm{d}\mathbf{r} \cdot \nabla\psi$ so

$$\frac{\mathrm{d}I}{\mathrm{d}\psi} = \frac{\mathrm{d}\mathbf{r} \cdot \nabla I}{\mathrm{d}\mathbf{r} \cdot \nabla\psi}. \tag{9.46}$$

Since ∇I is parallel to $\nabla\psi$, the derivative $\mathrm{d}I/\mathrm{d}\psi$ is always defined and so, in principle, may always be integrated. Thus, I must be a function of ψ and it is therefore always possible to write

$$\nabla I(\psi) = I'(\psi)\nabla\psi, \tag{9.47}$$

where prime means derivative with respect to the argument. The poloidal current can thus be expressed in terms of the poloidal flux function as

$$\mathbf{J}_{pol} = \frac{I'}{2\pi}\nabla\psi \times \nabla\phi. \tag{9.48}$$

Substitution for the currents and magnetic fields in Eq. (9.44) gives the expression

$$\begin{aligned} \nabla P &= \frac{I'}{2\pi}(\nabla\psi \times \nabla\phi) \times \frac{\mu_0 I}{2\pi}\nabla\phi - \nabla\phi\frac{r^2}{2\pi\mu_0}\nabla\cdot\left(\frac{1}{r^2}\nabla\psi\right) \times \frac{1}{2\pi}[\nabla\psi \times \nabla\phi] \\ &= -\left[\frac{\mu_0 I I'}{(2\pi r)^2} + \frac{1}{(2\pi)^2\mu_0}\nabla\cdot\left(\frac{1}{r^2}\nabla\psi\right)\right]\nabla\psi. \end{aligned} \tag{9.49}$$

This gives the important result that ∇P must also be parallel to $\nabla\psi$, which in turn implies $P = P(\psi)$ so $\nabla P = P'\nabla\psi$. Equation (9.49) now has a common vector factor $\nabla\psi$, which may be divided out, in which case the original vector equation reduces to the *scalar* equation

$$\nabla\cdot\left(\frac{1}{r^2}\nabla\psi\right) + 4\pi^2\mu_0 P' + \frac{\mu_0^2}{r^2}II' = 0. \tag{9.50}$$

This equation, known as the Grad–Shafranov equation (Grad and Rubin 1958, Shafranov 1966), has the peculiarity that ψ shows up as both an independent

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variable and as a dependent variable, i.e., there are both derivatives of ψ and derivatives with respect to ψ. Axisymmetry has made it possible to transform a three-dimensional vector equation into a one-dimensional scalar equation. It is not surprising that axisymmetry would transform a three-dimensional system into a two-dimensional system, but the transformation of a three-dimensional system into a one-dimensional system suggests more profound physics is involved than just geometrical simplification.

The Grad–Shafranov equation can be substituted back into Eq. (9.42) to give

$$\mathbf{J}_{tor} = \left(2\pi r^2 P' + \frac{\mu_0}{2\pi} I I'\right) \nabla\phi \tag{9.51}$$

so the total current can be expressed as

$$\mathbf{J} = \mathbf{J}_{pol} + \mathbf{J}_{tor} = \frac{I'}{2\pi} \nabla\psi \times \nabla\phi + \left(2\pi r^2 P' + \frac{\mu_0}{2\pi} I I'\right) \nabla\phi = 2\pi r^2 P' \nabla\phi + I'\mathbf{B}. \tag{9.52}$$

The last term is called the "force-free" current because it is parallel to the magnetic field and so provides no force. The first term on the right-hand side is the diamagnetic current.

The Grad–Shafranov equation is a nonlinear equation in ψ and, in general, cannot be solved analytically. It does not in itself determine the equilibrium because it involves three independent quantities, ψ, $P(\psi)$, and $I(\psi)$. Thus, use of the Grad–Shafranov equation involves specifying two of these functions and then solving for the third. Typically $P(\psi)$ and $I(\psi)$ are specified (either determined from other equations or from experimental data) and then the Grad–Shafranov equation is used to determine ψ.

Although the Grad–Shafranov equation must in general be solved numerically, there exists a limited number of analytic solutions. These can be used as idealized examples that demonstrate typical properties of axisymmetric equilibria. We shall now examine one such analytic solution, the Solov'ev solution (Solov'ev 1976).

The Solov'ev solution is obtained by invoking two assumptions: first, the pressure is assumed to be a linear function of ψ,

$$P = P_0 + \lambda\psi \tag{9.53}$$

and second, I is assumed to be constant within the plasma so

$$I' = 0. \tag{9.54}$$

The second assumption corresponds to having all the z-directed current flowing on the z axis, so that away from the z axis, the toroidal field is a vacuum field (cf. Eq. (9.35)). This arrangement is equivalent to having a zero-radius current-carrying wire going up the z axis acting as the sole source for the toroidal magnetic field. The region over which the Solev'ev solution applies excludes the z axis and so the problem is analogous to a central force problem where the source of the central force is a singularity at the origin. The excluded region of the z axis corresponds to the "hole in the doughnut" of a tokamak. In the more general case where I' is finite, the plasma is either diamagnetic (toroidal field is weaker than the vacuum field) or paramagnetic (toroidal field is stronger than the vacuum field).

Using the two simplifying assumptions provided by Eqs. (9.53) and (9.54), the Grad–Shafranov equation reduces to

$$r\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial \psi}{\partial r}\right)+\frac{\partial^2\psi}{\partial z^2}+4\pi^2 r^2\mu_0\lambda=0, \tag{9.55}$$

which has the exact solution

$$\psi(r,z)=\psi_0\frac{r^2}{r_0^4}\left(2r_0^2-r^2-4\alpha^2 z^2\right), \tag{9.56}$$

where $\psi_0, r_0,$ and α are constants; Eq. (9.56) is called the Solov'ev solution. Figure 9.11 shows a contour plot of ψ as a function of r/r_0 and z/z_0 for the case $\alpha=1$. Note that there are three distinct types of curves in Fig. 9.11, namely (i) open curves going to $z=\pm\infty$, (ii) concentric closed curves, and (iii) a single curve, called the separatrix, that separates the first two types of curves.

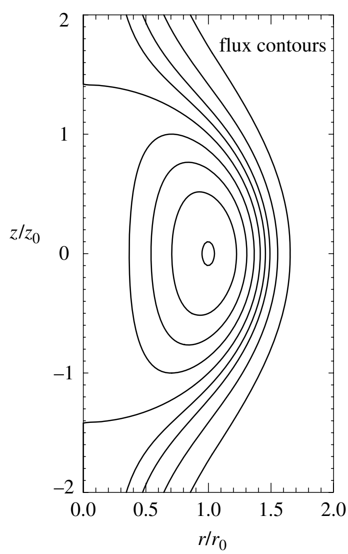

Fig. 9.11 Contours of constant flux of Solov'ev solution to Grad–Shafranov equation.

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Even though it is an idealization, the Solov'ev solution illustrates many important features of three-dimensional, static MHD equilibria. Let us now examine some of these features. The constants in Eq. (9.56) have been chosen so that $\psi = \psi_0$ at $r = r_0, z = 0$. The location $r = r_0, z = 0$ is called the *magnetic axis* because it is the axis linked by the closed contours of poloidal flux.

Let us temporarily assume ψ_0 is positive. Examination of Eq. (9.56) shows that if ψ is positive, then the quantity $2r_0^2 - r^2 - 4\alpha^2 z^2$ is positive and vice versa. For any negative value of ψ, Eq. (9.56) can be satisfied by making r very small and z very large. In particular, if r is infinitesimal then z must become infinite. Thus, all contours for which ψ is negative go to $z = \pm\infty$; these contours are the open or type (i) contours.

On the other hand, if ψ is positive then it has a maximum value of ψ_0, which occurs on the magnetic axis and if $0 < \psi < \psi_0$ then ψ must be located at some point outside the magnetic axis, but inside the curve $2r_0^2 - r^2 - 4\alpha^2 z^2 = 0$. Hence all contours of positive ψ must lie inside the curve $2r_0^2 - r^2 - 4\alpha^2 z^2 = 0$ and correspond to field lines that do not go to infinity. These contours are the closed or type (ii) contours, since they form closed curves in the r, z plane.

The contour separating the closed contours from the contours going to infinity is the *separatrix* and is given by the ellipse

$$r^2 + 4\alpha^2 z^2 = 2r_0^2. \tag{9.57}$$

The magnetic axis is a local maximum of ψ if ψ_0 is positive or a local minimum if ψ_0 is negative (the sign depends on the sense of the toroidal current). Thus, one can imagine that a hill (or valley) of poloidal flux exists with apex (or bottom) at the magnetic axis and in the vicinity of the apex (bottom) the contours of constant ψ are circles or ellipses enclosing the magnetic axis. The degree of ellipticity of these surfaces is determined by the value of α. These closed flux surfaces are a set of nested toroidal surfaces sharing the same magnetic axis. The projection of the total magnetic field lies in a flux surface since Eq. (9.34) shows

$$\mathbf{B} \cdot \nabla\psi = 0,$$

i.e., the magnetic field has no component in the direction normal to the flux surface.

Direct substitution of Eq. (9.56) into Eq. (9.55) gives

$$\lambda = \frac{2\psi_0}{\pi^2 r_0^4 \mu_0}\left(1 + \alpha^2\right) \tag{9.58}$$

so the pressure is

$$P(\psi) = P_0 + \frac{2\psi_0}{\pi^2 r_0^4 \mu_0}\left(1 + \alpha^2\right)\psi. \tag{9.59}$$

Pressure vanishes at the plasma edge so if ψ_{edge} is the poloidal flux at the edge, then the constant P_0 is determined and Eq. (9.59) becomes

$$P(\psi) = \frac{2\psi_0}{\pi^2 r_0^4 \mu_0}\left(1+\alpha^2\right)\left[\psi - \psi_{edge}\right]. \tag{9.60}$$

In the earlier discussion of single particle motion it was shown that particles were attached to flux surfaces. Particles starting on a particular flux surface remain on that flux surface, although they may travel to any part of that flux surface. The equilibrium prescribed by Eq. (9.56) is essentially a three-dimensional vortex with pressure peaking on the magnetic axis and then falling monotonically to zero at the edge.

This is a particularly simple solution to the Grad–Shafranov equation, but nevertheless demonstrates several important features, namely:

1. Three-dimensional equilibria can incorporate closed nested poloidal flux surfaces that are concentric about a magnetic axis. The magnetic axis is the local maximum or minimum of the poloidal flux ψ.
2. There can also be open flux surfaces; i.e., flux surfaces that go to infinity.
3. The flux surface separating closed and open flux surfaces is called a separatrix.
4. The total magnetic field projects into the flux surfaces since $\mathbf{B}\cdot\nabla\psi = 0$.
5. Finite pressure corresponds to a depression (or elevation) of the poloidal flux. The pressure maximum and flux extrema are located at the magnetic axis.
6. The poloidal magnetic field and associated poloidal flux ψ are responsible for plasma confinement and result from the toroidal current. Thus *toroidal* current is essential for confinement in an axisymmetric geometry. For a given $P(\psi)$ the toroidal magnetic field does not contribute to confinement if $I' = 0$. If I' is finite, plasma diamagnetism or paramagnetism will affect $P(\psi)$. Although a vacuum toroidal field cannot directly provide confinement, it can affect the rate of cross-field particle and energy diffusion and hence the functional form of $P(\psi)$. For example, the frequency of drift waves will be a function of the toroidal field and drift waves can cause an outward diffusion of plasma across flux surfaces, thereby affecting $P(\psi)$.
7. The poloidal flux surfaces are related to the surfaces of constant canonical angular momentum since $\mathbf{B}_{pol} = \nabla\times(\psi\nabla\phi/2\pi) = \nabla\times\left(\psi\hat{\phi}/2\pi r\right) = \nabla\times A_\phi\hat{\phi}$ implies $A_\phi = \psi/2\pi r$. Thus, the canonical angular momentum can be expressed as

$$\begin{aligned} p_\phi &= mr^2\dot{\phi} + qrA_\phi \\ &= mrv_\phi + \frac{q\psi}{2\pi}. \end{aligned} \tag{9.61}$$

Surfaces of constant canonical angular momentum correspond to surfaces of constant poloidal flux in the limit $m \to 0$. Because the system is toroidally symmetric (axisymmetric) the canonical angular momentum of each particle is a constant of the motion and so, to the extent that the particles can be approximated as having zero mass, particle trajectories are constrained to lie on surfaces of constant ψ. Since p_ϕ is a conserved quantity, the maximum excursion a finite-mass particle can make from a poloidal flux surface can be estimated by writing

$$\delta p_\phi = \delta\left(mrv_\phi + qrA_\phi\right) = 0. \tag{9.62}$$

Thus,

$$\frac{v_\phi}{r}\delta r + \delta v_\phi + \frac{q}{m}\delta r \frac{1}{r}\frac{\partial}{\partial r}\left(rA_\phi\right) = 0 \tag{9.63}$$

or

$$\delta r = -\frac{r\delta v_\phi}{v_\phi + r\omega_{c,pol}}, \tag{9.64}$$

where $\omega_{c,pol} = qB_{pol}/m$ is the cyclotron frequency measured using the poloidal magnetic field. Since v_ϕ is of the order of the thermal velocity v_T or smaller, if the poloidal field is sufficiently strong to make the poloidal Larmor radius $r_{lpol} = v_T/\omega_{c,pol}$ much smaller than the nominal configuration radius r, then the first term in the denominator may be dropped. Since δv_ϕ is also of the order of the thermal velocity or smaller, this gives the important result that *a particle cannot deviate from its initial flux surface by more than a poloidal Larmor radius*. Since poloidal field is produced by toroidal current, it is clear that particle confinement to axisymmetric flux surfaces requires the existence of toroidal current.

Tokamaks, reverse field pinches, spheromaks, and field reversed configurations are all magnetic confinement configurations having three-dimensional axisymmetric equilibria similar to this Solov'ev solution and all have a toroidal current producing a set of nested, closed poloidal flux surfaces that link a magnetic axis. Stellarators are non-axisymmetric configurations that have poloidal flux surfaces without toroidal currents; the advantage of current-free operation is offset by the complexity of non-axisymmetry.

9.9 Dynamic equilibria: flows

MHD-driven flows are relevant to arcs, magnetoplasmadynamic thrusters, electric currents in molten metals, structures on the solar corona, and astrophysical jets. The basic mechanism driving MHD flows will be discussed in Section 9.9.1 using the simplified assumptions of incompressibility and self-field only. The more general situation where the plasma is compressible and where there are external, applied magnetic fields in addition to the self-field will then be addressed in Section 9.9.2.

9.9.1 Incompressible plasma with self-field only

MHD-driven flows involve situations where $\nabla\times(\mathbf{J}\times\mathbf{B})$ is finite. This means the MHD force $\mathbf{J}\times\mathbf{B}$ is non-conservative, in which case its finite curl acts as a torque

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or equivalently as a source for hydrodynamic vorticity. When the MHD force is non-conservative a closed line integral $\oint d\mathbf{l} \cdot \mathbf{J} \times \mathbf{B}$ will be finite, whereas in contrast a closed line integral of a pressure gradient $\oint d\mathbf{l} \cdot \nabla P$ is always zero. Since a pressure gradient cannot balance the torque produced by a non-conservative $\mathbf{J} \times \mathbf{B}$, it is necessary to include a vorticity-damping term (viscosity term) to allow for the possibility of balancing this torque. Plasma viscosity is mainly due to ion–ion collisions and is typically very small. With the addition of viscosity, the MHD equation of motion becomes

$$\rho \left(\frac{\partial \mathbf{U}}{\partial t} + \mathbf{U} \cdot \nabla \mathbf{U} \right) = \mathbf{J} \times \mathbf{B} - \nabla P + \rho \upsilon \nabla^2 \mathbf{U}, \tag{9.65}$$

where υ is the kinematic viscosity. To focus attention on the generation, transport, and decay of vorticity, the following simplifying assumptions are made:

1. The motion is incompressible so that $\rho = const.$
   This is an excellent assumption for a molten metal, but questionable for a plasma. However, the assumption could be at least locally reasonable for a plasma if the region of vorticity generation is smaller in scale than regions of compression or rarefaction or is spatially distinct from these regions. The consequences of compressibility will be discussed later.
2. The system is cylindrically symmetric.
   A cylindrical coordinate system r, ϕ, z can then be used and ϕ is ignorable. This geometry corresponds to magnetohydrodynamic thrusters and arcs.
3. The flow velocities $\mathbf{U}$ and the current density $\mathbf{J}$ are in the poloidal direction (i.e., r and z directions). Restricting the current to be poloidal means that the magnetic field is purely toroidal (see Fig. 9.10).

The most general poloidal velocity for a constant-density, incompressible fluid has the form

$$\mathbf{U} = \frac{1}{2\pi} \nabla \mathcal{F} \times \nabla \phi, \tag{9.66}$$

where $\mathcal{F}(r, z)$ is the flux of fluid through a circle of radius r at location z. Note the analogy to the poloidal magnetic flux function $\psi(r, z)$ used in Eqs. (3.147) and (9.34). Since $\mathbf{U}$ lies entirely in the $r - z$ plane and since the system is axisymmetric, the curl of $\mathbf{U}$ will be in the ϕ direction. It is thus useful to define a scalar cylindrical "vorticity" $\chi = r\hat{\phi} \cdot \nabla \times \mathbf{U}$; this definition differs slightly from the conventional definition of vorticity (curl of velocity) because it includes an extra factor of r. This slight change of definition is essentially a matter of semantics, but makes the algebra more transparent.

Since $\nabla \phi = \hat{\phi}/r$ it is seen that

$$\nabla \times \mathbf{U} = \frac{\hat{\phi}}{r} r\hat{\phi} \cdot \nabla \times \mathbf{U} = \chi \nabla \phi \tag{9.67}$$

and so

$$\nabla\times\left(\frac{1}{2\pi}\nabla\mathcal{F}\times\nabla\phi\right)=\chi\nabla\phi. \tag{9.68}$$

Dotting Eq. (9.68) with $\nabla\phi$ and using the vector identity $\nabla\phi\cdot\nabla\times\mathbf{Q}=\nabla\cdot(\mathbf{Q}\times\nabla\phi)$ gives

$$\nabla\cdot\left(\frac{1}{2\pi}\left(\nabla\mathcal{F}\times\nabla\phi\right)\times\nabla\phi\right)=\frac{\chi}{r^2} \tag{9.69}$$

or

$$r^2\nabla\cdot\left(\frac{1}{r^2}\nabla\mathcal{F}\right)=-2\pi\chi, \tag{9.70}$$

which is a Poisson-like relation between the vorticity and the stream function. The vorticity plays the role of the source (charge-density) and the stream function plays the role of the potential function (electrostatic potential). However, the elliptic operator is not exactly a Laplacian and when expanded has the form

$$r^2\nabla\cdot\left(\frac{1}{r^2}\nabla\mathcal{F}\right)=r\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial\mathcal{F}}{\partial r}\right)+\frac{\partial^2\mathcal{F}}{\partial z^2}, \tag{9.71}$$

which is just the operator in the Grad–Shafranov equation.

Since the current was assumed to be purely poloidal, the magnetic field must be purely toroidal and so can be written as

$$\mathbf{B}=\frac{\mu_0}{2\pi}I\nabla\phi, \tag{9.72}$$

where I is the current through a circle of radius r at location z. This is consistent with the integral form of Ampère's law, $\oint\mathbf{B}\cdot d\mathbf{l}=\mu_0 I$. The curl of Eq. (9.72) gives

$$\nabla\times\mathbf{B}=\frac{\mu_0}{2\pi}\nabla I\times\nabla\phi \tag{9.73}$$

showing that I acts like a stream-function for the magnetic field.

The magnetic force is therefore

$$\mathbf{J}\times\mathbf{B}=\frac{1}{2\pi}\left(\nabla I\times\nabla\phi\right)\times\frac{\mu_0}{2\pi}I\nabla\phi=-\frac{\mu_0}{(2\pi r)^2}\nabla\left(\frac{I^2}{2}\right). \tag{9.74}$$

It is seen that there is an axial force if I^2 depends on z; this is the essential condition that produces axial flows in MHD arcs (Maecker 1955) and plasma guns (Marshall 1960).

Using the identities $\nabla U^2/2=\mathbf{U}\cdot\nabla\mathbf{U}+\mathbf{U}\times\nabla\times\mathbf{U}$ and $\nabla\times\nabla\times\mathbf{U}=-\nabla^2\mathbf{U}=\nabla\chi\times\nabla\phi$ the equation of motion Eq. (9.65) becomes

$$\rho\left(\frac{\partial\mathbf{U}}{\partial t}+\nabla\frac{U^2}{2}-\mathbf{U}\times\chi\nabla\phi\right)=-\frac{\mu_0}{(2\pi r)^2}\nabla\left(\frac{I^2}{2}\right)-\nabla P-\rho\upsilon\nabla\chi\times\nabla\phi \tag{9.75}$$

or

$$\rho\left(\frac{\partial \mathbf{U}}{\partial t}+\nabla\frac{U^2}{2}-\mathbf{U}\times\chi\nabla\phi\right)=-\frac{\mu_0}{(2\pi)^2}\nabla\left(\frac{I^2}{2r^2}\right)+\frac{\mu_0 I^2}{(2\pi)^2}\nabla\left(\frac{1}{2r^2}\right)$$
$$-\nabla P-\rho\upsilon\nabla\chi\times\nabla\phi. \tag{9.76}$$

Every term in this equation is either a gradient of a scalar or else can be expressed as a cross-product involving $\nabla\phi$. The equation can thus be regrouped as

$$\nabla\left(\frac{\rho U^2}{2}+\frac{\mu_0}{(2\pi)^2}\frac{I^2}{2r^2}+P\right)+\left(\frac{\rho}{2\pi}\nabla\frac{\partial\mathcal{F}}{\partial t}-\rho\mathbf{U}\chi-\frac{\mu_0 I^2}{(2\pi r)^2}\hat{z}+\rho\upsilon\nabla\chi\right)\times\nabla\phi=0. \tag{9.77}$$

Similarly, the MHD Ohm's law $\mathbf{E}+\mathbf{U}\times\mathbf{B}=\eta\mathbf{J}$ can be written as

$$-\frac{\partial\mathbf{A}}{\partial t}-\nabla V+\mathbf{U}\times\frac{\mu_0}{2\pi}I\nabla\phi=\frac{\eta}{2\pi}\nabla I\times\nabla\phi, \tag{9.78}$$

where V is the electrostatic potential. Since $\mathbf{B}$ is purely toroidal, $\mathbf{A}$ is poloidal and so is orthogonal to $\nabla\phi$. Thus, both the equation of motion and Ohm's law are equations of the form

$$\nabla g+\mathbf{Q}\times\nabla\phi=0, \tag{9.79}$$

which is the most general form of an axisymmetric partial differential equation involving a potential. Two distinct scalar partial differential equations can be extracted from Eq. (9.79) by (i) operating with $\nabla\phi\cdot\nabla\times$ and (ii) taking the divergence. Doing the former it is seen that Eq. (9.79) becomes

$$\nabla\phi\cdot\nabla\times(\mathbf{Q}\times\nabla\phi)=0 \tag{9.80}$$

or

$$\nabla\cdot\left(\frac{1}{r^2}\mathbf{Q}_{pol}\right)=0. \tag{9.81}$$

Applying this procedure to Eq. (9.77) gives

$$\frac{\rho}{2\pi}\frac{\partial}{\partial t}\left(\nabla\cdot\frac{1}{r^2}\nabla\mathcal{F}\right)-\rho\nabla\cdot\left(\frac{1}{r^2}\mathbf{U}\chi\right)-\nabla\cdot\left(\frac{\mu_0 I^2}{\left(2\pi r^2\right)^2}\hat{z}\right)+\rho\upsilon\nabla\cdot\left(\frac{1}{r^2}\nabla\chi\right)=0 \tag{9.82}$$

or

$$\frac{\partial}{\partial t}\left(\frac{\chi}{r^2}\right)+\nabla\cdot\left(\mathbf{U}\frac{\chi}{r^2}\right)=-\frac{\mu_0}{4\pi^2\rho r^4}\frac{\partial I^2}{\partial z}+\upsilon\nabla\cdot\left(\frac{1}{r^2}\nabla\chi\right). \tag{9.83}$$

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Since $\nabla\cdot\mathbf{U}=0$, this can also be written as

$$\frac{\partial}{\partial t}\left(\frac{\chi}{r^2}\right)+\mathbf{U}\cdot\nabla\left(\frac{\chi}{r^2}\right)=-\frac{\mu_0}{4\pi^2\rho r^4}\frac{\partial I^2}{\partial z}+\upsilon\nabla\cdot\frac{1}{r^2}\nabla\chi, \tag{9.84}$$

which shows that if there is no viscosity and if $\partial I^2/\partial z=0$, then the scaled vorticity χ/r^2 convects with the fluid; i.e., is frozen into the fluid. The viscous term on the right-hand side describes a diffusive-like dissipation of vorticity. The remaining term $r^{-4}\partial I^2/\partial z$ acts as a *vorticity source* and is finite only if I^2 is non-uniform in the z direction. The vorticity source has a strong r^{-4} weighting factor so axial non-uniformities of I near $r=0$ dominate. Positive χ corresponds to a clockwise rotation in the $r-z$ plane. If I^2 is an increasing function of z then the source term is negative, implying that a counterclockwise vortex is generated and vice versa. Suppose as shown in Fig. 9.12 that the current channel becomes wider with increasing z, corresponding to a fanning out of the current with increasing z. In this case $\partial I^2/\partial z$ will be negative and a clockwise vortex will be generated. Fluid will flow radially inwards at small z, then flow vertically upwards, and finally radially outwards at large z. The fluid flow produced by the vorticity source will convect the vorticity along with the flow until it is dissipated by viscosity.

Operating on Ohm's law, Eq. (9.78), with $\nabla\phi\cdot\nabla\times$, gives

$$\frac{\partial}{\partial t}\left(\frac{I}{r^2}\right)+\mathbf{U}\cdot\nabla\left(\frac{I}{r^2}\right)=\frac{\eta}{\mu_0}\nabla\cdot\left(\frac{1}{r^2}\nabla I\right) \tag{9.85}$$

showing that I/r^2 is similarly convected with the fluid and also has a diffusive-like term, this time with coefficient η/μ_0.

Thus, the system of equations can be summarized as (Bellan 1992)

$$\frac{\partial}{\partial t}\left(\frac{\chi}{r^2}\right)+\mathbf{U}\cdot\nabla\left(\frac{\chi}{r^2}\right)=\upsilon\nabla\cdot\frac{1}{r^2}\nabla\chi-\frac{\mu_0}{4\pi^2\rho r^4}\frac{\partial I^2}{\partial z} \tag{9.86}$$

$$\frac{\partial}{\partial t}\left(\frac{I}{r^2}\right)+\mathbf{U}\cdot\nabla\left(\frac{I}{r^2}\right)=\frac{\eta}{\mu_0}\nabla\cdot\left(\frac{1}{r^2}\nabla I\right) \tag{9.87}$$

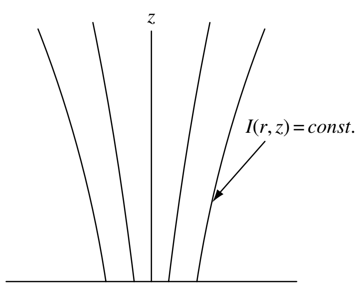

Fig. 9.12 Current channel $I(r,z)$.

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$$r^2\nabla\cdot\left(\frac{1}{r^2}\nabla\mathcal{F}\right)=-2\pi\chi \tag{9.88}$$

$$\mathbf{U}=\frac{1}{2\pi}\nabla\mathcal{F}\times\nabla\phi. \tag{9.89}$$

This system cannot be solved analytically, but its general behavior can be described in a qualitative manner. The system involves three scalar variables, $\chi, \mathcal{F}$, and I, all of which are functions of r and z. Boundary conditions must be specified in order to have a well-posed problem. If a recirculating flow is driven, the fluid flux $\mathcal{F}$ will be zero on the boundary whereas I will be finite on electrodes at the boundary. In particular, the current I will typically flow from the anode into the plasma and then from the plasma into the cathode. The vorticity χ will usually have the boundary condition of vanishing on the bounding surface.

Initially, there is no flow so $\mathcal{F}$ is zero everywhere. An initial solution of Eq. (9.87) in this no-flow situation will establish a current profile between the two electrodes. For any reasonable situation, this $I(r,z)$ profile will have $\partial I^2/\partial z\neq 0$ so that a vorticity source will be created. The source will be quite localized because of the r^{-4} coefficient. Once some vorticity χ is created as determined by Eq. (9.86), this vorticity acts as a source term for the fluid flux $\mathcal{F}$ in Eq. (9.88), and so a finite $\mathcal{F}$ will be developed. Thus a flow $\mathbf{U}(r,z)$ will be created as specified by Eq. (9.89), and this flow will convect both χ/r^2 and I/r^2.

A good analogy is to think of the $r^{-4}\partial I^2/\partial z$ term as constituting a toroidally symmetric centrifugal pump, which accelerates fluid radially from large to small r. This radial acceleration takes place at z locations where the current channel radius is constricted. The pump then accelerates the ingested fluid up or down the z axis, in a direction away from the current constriction. This vortex generation can also be seen by drawing vectors showing the magnitude and orientation of the $\mathbf{J}\times\mathbf{B}$ force in the vicinity of a current constriction. It is seen that the $\mathbf{J}\times\mathbf{B}$ force is non-conservative and provides a centrifugal pumping as described above.

An arc or magnetoplasmadynamic thruster can thus be construed as a pump that draws fluid radially inward towards the smaller radius electrode, accelerates this fluid to high velocity, and then expels the accelerated fluid in the axial direction in a manner whereby the accelerated fluid shoots axially from the smaller electrode towards the larger radius electrode. The sign of the current does not matter, since the pumping action depends only on the z derivative of I^2. If the equations are put in dimensionless form, it is seen that the characteristic flow velocity is of the order of the Alfvén velocity.

Once $\chi, \mathcal{F}$, and I have been determined, it is possible to determine the pressure and electrostatic potential profiles. This is done by taking the divergence of

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Eq. (9.79) to obtain

$$\nabla^2 g = -\nabla\phi \cdot \nabla \times \mathbf{Q}. \tag{9.90}$$

The pressure and electrostatic potential are contained within the g term, whereas the $\mathbf{Q}$ term involves only χ, $\mathcal{F}$, and I or functions of χ, $\mathcal{F}$, and I (e.g., the velocity). Thus, the right-hand side of Eq. (9.90) is known and so can be considered as the source for a Poisson-like equation for the left-hand side. For the equation of motion

$$g_{motion} = \frac{\rho U^2}{2} + \frac{\mu_0}{(2\pi)^2}\frac{I^2}{2r^2} + P \tag{9.91}$$

so that

$$P = g_{motion} - \frac{\rho U^2}{2} - \frac{\mu_0}{(2\pi)^2}\frac{I^2}{2r^2}. \tag{9.92}$$

Regions where g_{motion} is constant satisfy an extended form of the Bernoulli theorem, namely

$$\frac{\rho U^2}{2} + \frac{B^2}{2\mu_0} + P = const. \tag{9.93}$$

Similarly, taking the divergence of Eq. (9.78) gives an equation for the electrostatic potential (using Coulomb gauge so that $\nabla \cdot \mathbf{A} = 0$). Thus, a complete solution for incompressible flow is obtained by first solving for χ, $\mathcal{F}$, and I using prescribed boundary conditions, and then solving for the pressure and electrostatic potential.

9.9.2 Compressible plasma and applied poloidal field

The previous discussion assumed that the plasma was incompressible and that the magnetic field was purely toroidal (i.e., generated by the prescribed poloidal current). The more general situation involves having a pre-existing poloidal magnetic field such as would be generated by external coils and also allowing for plasma compressibility. This situation will now be discussed qualitatively making reference to the sketch provided in Fig. 9.13 (a more thorough discussion is provided in Bellan (2003).

Consider the initial situation shown in Fig. 9.13(a) in which a plasma is immersed in an axisymmetric vacuum poloidal field $\psi(r, z)$ with a poloidal current $I(r, z)$. Because ψ is assumed to correspond to a vacuum field, it is produced by external coils with toroidal currents. There are thus no toroidal currents in the plasma volume under consideration (i.e., in the region described by Fig. 9.13(a))

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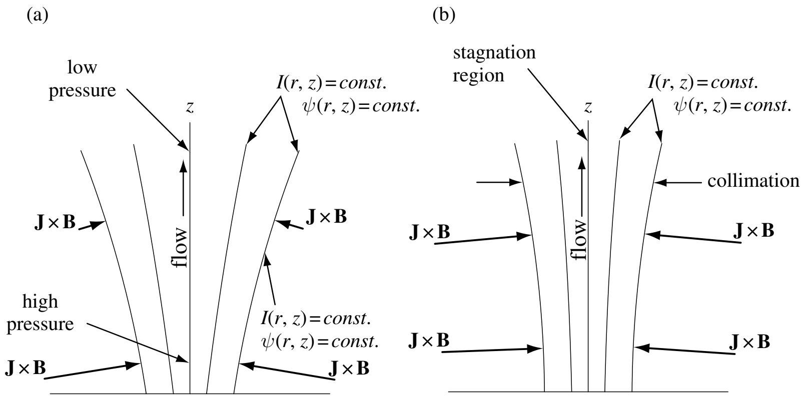

Fig. 9.13 (a) Current channel $I(r, z)$, which has same contours as poloidal flux surfaces $\psi(r, z)$ so that ∇I is parallel to $\nabla\psi$ in order to have no toroidal acceleration. The $\mathbf{J}\times\mathbf{B}$ force (shown by thick arrows) is larger at the bottom and canted giving both a higher on-axis pressure at the bottom and an axial upwards flow. (b) If the flow stagnates (slows down) for some reason, then mass accumulates at the top. The frozen-in convected toroidal flux also accumulates, which thus increases B_ϕ and so forces r to decrease since Ampère's law gives $2\pi B_\phi r = I(r, z)$, which is constant on a surface of constant $I(r, z)$. The upward-moving, stagnating plasma flow with embedded toroidal flux acts somewhat like a zipper for the surfaces of constant $I(r, z)$ and $\psi(r, z)$.

and so using Eq. (9.42) it is seen that $\psi(r, z)$ satisfies $\nabla\cdot\left(r^{-2}\nabla\psi\right) = 0$. The poloidal current $I(r, z)$ has an associated poloidal current density

$$\mathbf{J}_{pol} = \frac{1}{2\pi}\nabla I\times\nabla\phi \tag{9.94}$$

and an associated toroidal magnetic field

$$B_\phi = \frac{\mu_0 I(r, z)}{2\pi r}. \tag{9.95}$$

The poloidal flux $\psi(r, z)$ has an associated poloidal magnetic field

$$\mathbf{B}_{pol} = \frac{1}{2\pi}\nabla\psi\times\nabla\phi. \tag{9.96}$$

If $\mathbf{J}_{pol}$ is parallel to $\mathbf{B}_{pol}$ then ∇I would be parallel to $\nabla\psi$ so that constant $I(r, z)$ surfaces coincide with constant $\psi(r, z)$ surfaces as indicated in the figure. On the other hand, if $\mathbf{J}_{pol}$ were not parallel to $\mathbf{B}_{pol}$, these surfaces would not be coincident and there would be a force $\mathbf{J}_{pol}\times\mathbf{B}_{pol}$ in the toroidal direction, which would tend to cause a change in the toroidal velocity, i.e., an acceleration or deceleration of U_ϕ. We argue that $\mathbf{J}_{pol}\times\mathbf{B}_{pol}$ must be a transient force with zero time average because any current that flows perpendicular to the poloidal flux surfaces must be

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the polarization current as given by Eq. (3.145). This polarization current results from the time derivative of the electric field perpendicular to the poloidal flux surfaces and this electric field in turn is proportional to the time rate of change of the poloidal current as determined from Faraday's law. Thus, once the current is in steady state, it is necessary to have $I = I(\psi)$ and so the surfaces of constant $I(r, z)$ in Fig. 9.12 can also be considered as surfaces of constant $\psi(r, z)$. Another way to see this is to realize that a steady current perpendicular to flux surfaces would cause a continuous electrostatic charging of the flux surfaces, which would then violate the stipulation that the plasma is quasi-neutral.

The initial magnetic force is thus the same as for the situation of a purely toroidal magnetic field since

$$\mathbf{J} \times \mathbf{B} = \underbrace{\mathbf{J}_{tor}}_{\text{zero}} \times \mathbf{B}_{pol} + \mathbf{J}_{pol} \times \mathbf{B}_{tor}. \tag{9.97}$$

The remaining force $\mathbf{J}_{pol} \times \mathbf{B}_{tor} = \left(J_r\hat{r} + J_z\hat{z}\right) \times B_\phi\hat{\phi} = J_rB_\phi\hat{z} - J_zB_\phi\hat{r}$ has components in both the r and z directions. If the plasma axial length is much larger than its radius (i.e., it is long and skinny) then the plasma will develop a local radial pressure balance

$$\left(-\nabla P + \mathbf{J}_{pol} \times \mathbf{B}_{tor}\right) \cdot \hat{r} = 0. \tag{9.98}$$

However, this local radial pressure balance precludes the possibility of an axial pressure balance if $\partial\psi/\partial z \neq 0$ because, as discussed earlier, ∇P cannot balance $\mathbf{J} \times \mathbf{B}$ if the latter has a finite curl. Suppose that the radius of the current channel and the poloidal flux are both given by $a = a(z)$ and, furthermore, since $I = I(\psi)$ let us assume a simple linear dependence where $\mu_0 I(r, z) = \lambda\psi(r, z)$. Thus, ∇I is parallel to $\nabla\psi$ and I is just the current per flux. We assume a parabolic poloidal flux for $r \leq a(z)$, namely

$$\frac{\psi(r, z)}{\psi_0} = \left(\frac{r}{a(z)}\right)^2, \tag{9.99}$$

where ψ_0 is the flux at $r = a$; this is the simplest allowed form for the flux (if ψ depended linearly on r, then, as noted in the discussion of Eq.(3.155), there would be infinite fields at $r = 0$). The local radial pressure balance is essentially a local version of the Bennett pinch relation

$$\begin{aligned} \frac{\partial P}{\partial r} &= -J_z B_\phi \\ &= -\frac{\lambda^2}{8\pi^2\mu_0 r^2}\frac{\partial\psi^2}{\partial r}. \end{aligned} \tag{9.100}$$

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On substitution of Eq. (9.99) this becomes

$$\begin{aligned}\frac{\partial P}{\partial r} &= -\frac{\mu_0 \lambda^2 \psi_0^2}{8\pi^2 r^2}\frac{\partial}{\partial r}\left(\frac{r}{a(z)}\right)^4 \\ &= -\frac{\lambda^2 \psi_0^2 r}{2\pi^2 \mu_0 a^4}.\end{aligned} \tag{9.101}$$

Integrating and using the boundary condition that $P = 0$ at $r = a$ gives

$$P(r) = \frac{\lambda^2 \psi_0^2}{4\pi^2 \mu_0 a^2}\left(1 - \frac{r^2}{a^2}\right), \tag{9.102}$$

which shows that the on-axis pressure is larger where $a(z)$ is smaller as indicated in Fig. 9.13(a). The axial component of the equation of motion is thus

$$\begin{aligned}F_z &= \hat{z}\cdot\left(-\nabla P + \mathbf{J}_{pol}\times\mathbf{B}_{tor}\right) \\ &= -\frac{\partial P}{\partial z} + J_r B_\phi \\ &= -\frac{\lambda^2\psi_0^2}{4\pi^2\mu_0}\frac{\partial}{\partial z}\left(\frac{1}{a^2} - \frac{r^2}{a^4}\right) - \frac{1}{\mu_0}B_\phi\frac{\partial B_\phi}{\partial z} \\ &= -\frac{\lambda^2\psi_0^2}{\pi^2\mu_0}\left(-\frac{1}{2a^3} + \frac{r^2}{a^5}\right)\frac{\partial a}{\partial z} - \frac{1}{2\mu_0}\frac{\partial}{\partial z}\left(\frac{\lambda\psi}{2\pi r}\right)^2 \\ &= \frac{\lambda^2\psi_0^2}{\pi^2 a^3\mu_0}\left(\frac{1}{2} - \frac{r^2}{a^2}\right)\frac{\partial a}{\partial z} - \frac{\lambda^2\psi_0^2}{8\pi^2 r^2\mu_0}\frac{\partial}{\partial z}\left(\frac{r}{a}\right)^4 \\ &= \frac{\mu_0 I_0^2}{2\pi^2 a^3}\left(1 - \frac{r^2}{a^2}\right)\frac{\partial a}{\partial z},\end{aligned} \tag{9.103}$$

which shows there is an axial force that is maximum on the z axis. This force is proportional to the current flowing along the flux tube and to the axial non-uniformity of the flux tube, i.e., to $\partial a/\partial z$. The axial force accelerates plasma away from regions where a is small to regions where a is large.

If the flow stagnates, i.e., is such that the axial velocity is non-uniform so that $\nabla\cdot\mathbf{U}$ is negative, then there will be a net inflow of matter into the stagnation region and hence an increase of mass density at the locations where $\nabla\cdot\mathbf{U}$ is negative. Because magnetic flux is frozen to the plasma, there will be a corresponding accumulation of toroidal magnetic flux at the locations where $\nabla\cdot\mathbf{U}$ is negative. If U_ϕ is zero (as assumed on the basis of there being no steady toroidal acceleration and zero initial toroidal velocity), then the accumulation of toroidal flux will

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increase the toroidal field. This can be seen by considering the toroidal component of the induction equation

$$\frac{\partial B_\phi}{\partial t} = r\mathbf{B}_{pol} \cdot \nabla \left(\frac{U_\phi}{r}\right) - r\mathbf{U}_{pol} \cdot \nabla \left(\frac{B_\phi}{r}\right) - B_\phi \nabla \cdot \mathbf{U}_{pol} \tag{9.104}$$

or

$$\frac{DB_\phi}{Dt} = -B_\phi \nabla \cdot \mathbf{U}_{pol}, \tag{9.105}$$

where $D/DT = \partial/\partial t + \mathbf{U} \cdot \nabla$ and $U_\phi = 0$ has been assumed. Thus, it is seen that B_ϕ will increase in the frame of the plasma at locations where $\nabla \cdot \mathbf{U}_{pol}$ is negative. However, Ampère's law gives $\mu_0 I = 2\pi r B_\phi$ and since $I = I_0$ at the outer radius of the current channel, it is seen that if B_ϕ increases, then the current channel radius has to decrease in order to keep rB_ϕ fixed. The result is that stagnation of the flow tends to collimate the flux tube, i.e., make it axially uniform as shown in Fig. 9.13(b). The plasma accelerated upwards in Fig. 9.12 will then squeeze together the ψ and thus I surfaces so that these surfaces will become vertical lines; very roughly, stagnation of an upward moving plasma in Fig. 9.12 can be imagined as a sort of "zipper," which collimates the flux surfaces. Collimation is often seen in current-carrying magnetic flux tubes in laboratory experiments (e.g., see Hansen and Bellan (2001)) and is an important property of solar coronal loops (Klimchuk 2000) and of astrophysical jets (Livio 1999). The MHD pumping prescribed by Eq. (9.103) and its association with flux tube collimation has recently been observed in laboratory experiments (You, Yun, and Bellan 2005).

9.10 Assignments

1. Show that if a Bennett pinch equilibrium has J_z independent of r, then the azimuthal magnetic field is of the form $B_\theta(r) = B_\theta(a)r/a$. Then show that if the temperature is uniform, the pressure will be of the form $P = P_0(1 - r^2/a^2)$, where $P_0 = \mu_0 I^2/4\pi^2 a^2$. Show that this result is consistent with Eq. (9.29).
2. Show it is impossible to confine a spherically symmetric pressure profile using magnetic forces alone and therefore show that three-dimensional magnetostatic equilibria cannot be found for arbitrarily specified pressure profiles. To do this, suppose there exists a magnetic field satisfying $\mathbf{J} \times \mathbf{B} = \nabla P$, where $P = P(r)$ and r is the radius in spherical coordinates $\{r, \theta, \phi\}$. Then make the following analysis:
   (a) Show that neither $\mathbf{J}$ nor $\mathbf{B}$ can have a component in the $\hat{r}$ direction. Hint: assume B_r is finite and write out the three components of $\mathbf{J} \times \mathbf{B} = \nabla P$ in spherical coordinates. By eliminating either J_r or B_r from the equations resulting from the θ and ϕ components, obtain equations of the form $J_r \partial P/\partial r = 0$ and $B_r \partial P/\partial r = 0$. Then use the fact that $\partial P/\partial r$ is finite to develop a contradiction regarding the original assumption regarding J_r or B_r.

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(b) By calculating J_θ and J_ϕ from Ampère's law (use the curl in spherical coordinates as given on p. 592), show that $\mathbf{J}\times\mathbf{B}=\nabla P$ implies $B_\theta^2+B_\phi^2=B^2(r)$. Argue that this means that the magnitude of $\mathbf{B}=B_\theta\hat{\theta}+B_\phi\hat{\phi}$ must be independent of θ and ϕ and therefore B_θ and B_ϕ must be of the form $B_\theta=B(r)\sin[\eta(\theta,\phi,r)]$ and $B_\phi=B(r)\cos[\eta(\theta,\phi,r)]$, where η is some arbitrary function.

(c) Using $J_r=0$, $B_r=0$, and $\nabla\cdot\mathbf{B}=0$ (see p. 591 for the divergence in spherical polar coordinates) derive a pair of coupled equations for $\partial\eta/\partial\theta$ and $\partial\eta/\partial\phi$. Solve these coupled equations for $\partial\eta/\partial\theta$ and $\partial\eta/\partial\phi$ using the method of determinants (the solutions for $\partial\eta/\partial\theta$ and $\partial\eta/\partial\phi$ should be very simple). Integrate the $\partial\eta/\partial\phi$ equation to obtain η and show that this solution for η is inconsistent with the requirements of the solution for $\partial\eta/\partial\theta$, thereby demonstrating that the original assumption of existence of a spherically symmetric equilibrium must be incorrect.

3. Using numerical methods solve for the motion of charged particles in the Solov'ev field, $\mathbf{B}=\nabla\psi\times\nabla\phi$, where $\psi=B_0r^2(2a^2-r^2-4(\alpha z)^2)/a^4$ and B_0, a, and α are constants. Plot the surfaces of constant ψ and show there are open and closed surfaces. Select appropriate characteristic times, lengths, and velocities and choose appropriate time steps, initial conditions, and graph windows. Plot the x, y plane and the x, z plane. What interesting features are observed in the orbits? How do they relate to the flux surfaces $\psi=$ constant? What can be said about flux conservation?

4. Grad–Shafranov equation for an axisymmetric current-carrying magnetic flux tube. Because the Grad–Shafranov equation describes axisymmetric equilibria, in addition to describing axisymmetric configurations with closed flux surfaces (e.g., tokamaks), it can also be used to characterize an axisymmetric magnetic flux tube, i.e., a configuration with axisymmetric open flux surfaces that may or may not have an axially uniform cross-sectional area. Suppose that $\psi=\psi(r,z)$ and that ψ is a monotonically increasing function of r. The poloidal flux function thus defines the shape of an axisymmetric flux tube; in particular, the flux tube will be axially uniform only in the special case where ψ does not depend on z.

   Assume both the current I and pressure P are linear functions of ψ so

$$\mu_0 I=\lambda\psi$$
$$P=P_0(1-\psi/\psi_0),$$

where ψ_0 is the flux surface on which P vanishes and P_0 is the pressure on the z axis. Show that the Grad–Shafranov equation can then be written in the form

$$r^2\nabla\cdot\left(\frac{1}{r^2}\nabla\bar{\psi}\right)+\mu_0^2\lambda^2\bar{\psi}=\frac{4\pi^2r^2\mu_0P_0}{\psi_0^2},$$

where $\bar{\psi}=\psi(r,z)/\psi_0$. Show that if

$$P_0=\frac{\mu_0^2\lambda^2\psi_0^2}{4\pi^2\mu_0a^2},$$

where a is the radius at which the pressure vanishes at $z=0$, the flux tube must be axially uniform, i.e., ψ cannot have any z dependence. Express this condition in terms

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of the pitch angle of the magnetic field B_ϕ/B_z measured at $r=a$. Hint: note that the Grad–Shafranov equation has both a homogeneous part (left-hand side) and an inhomogeneous part (right-hand side). Show that $\bar{\psi}=r^2/a^2$ is a particular solution and then consider the form of the general solution (homogenous plus inhomogeneous solution).

5. Lawson Criterion. Of the many possible thermonuclear reactions, the deuterium-tritium (DT) reaction stands out as being the most feasible because it has the largest reaction cross-section at accessible temperatures. The DT reaction has the form

$$\mathrm{D}+\mathrm{T}\longrightarrow \mathrm{n}+\mathrm{He}^4+17.6\,\mathrm{MeV}.$$

The output energy consists of 3.5 MeV neutron kinetic energy and 14.1 MeV alpha particle kinetic energy. Fusion reaction cross-sections have an extreme dependence on the impact energy of reacting ions because Coulomb repulsion makes it energetically very difficult for two ions to approach one another. In order for a controlled fusion reaction to be economically useful, more energy must be generated than invested. The break-even condition is determined by equating the energy invested to the energy harvested from the fusion reaction. If the reaction dies prematurely then there is insufficient return on the investment. Thus, the lifetime of the fusion-producing plasma is a critical parameter and is characterized by the so-called *energy confinement* time τ_E of the configuration, a measure of the insulating capability of the plasma confinement device. Let all temperatures and energies be measured in kilovolts (keV) so the Boltzmann constant becomes $\kappa=1.6\times10^{-16}$ and consider a volume V of reacting deuterium, tritium, and associated neutralizing electrons (the plasma needs to be overall neutral, otherwise there would be enormous electrostatic forces).

(a) Show that if the electrons and ions have the same temperature T, the energy required to heat this volume is $E_{in}\sim 3n_eV\kappa T$, where n_e is the electron density and equal densities of deuterium and tritium are assumed so that $n_D=n_T=n_e/2$. (Hint: the mean energy per degree of freedom is $\kappa T/2$ so the mean energy of a particle moving in three-dimensional space is $3\kappa T/2$).

(b) What is the fusion reaction rate R for a single incident ion of velocity v passing through stationary target ions where $\sigma(v)$ is the fusion reaction rate cross-section and n_t is the density of the target ions (assumed stationary). If $\langle\sigma(v)v\rangle$ denotes the velocity average of $\sigma(v)v$, what is the velocity-averaged reaction rate?

(c) Assume that the reaction of hot tritium with hot deuterium can be approximated as the sum of hot deuterium interacting with stationary tritium and hot tritium interacting with stationary deuterium. Show that the fusion output power in the volume is

$$P=n_Dn_T\,\langle\sigma(v)v\rangle\,V\,E_{reaction},$$

where $E_{reaction}=17.6\times10^3$ keV.

(d) Show that the ratio Q of the fusion output energy to the input energy satisfies the relation

$$n_e\tau_E\sim\frac{6.8\times10^{-4}T_{keV}}{\langle\sigma(v)v\rangle}Q.$$

At break-even $Q=1$.

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(e) For a thermal ion distribution and the experimentally measured reaction cross-section $\sigma(v)$, the velocity-averaged rate integral has the values tabulated below:

Temperature in keV	$\langle\sigma(v)v\rangle$ in m^3/s	$n_e\tau_E$
1	5.5×10^{-27}	
2	2.6×10^{-25}	
5	1.3×10^{-23}	
10	1.1×10^{-22}	
20	4.2×10^{-22}	
50	8.7×10^{-22}	
100	8.5×10^{-22}	

Calculate the $n_e\tau_E$ required for break-even at the listed temperatures (i.e., fill in the last column).

(f) The Joint European Tokamak (JET) located in Culham, UK, (in operation during the latter twentieth and early twenty first century) has a plasma volume of approximately 80 cubic meters. Suppose JET has an energy confinement time of $\tau_E \sim 0.3$ s at $n_e \sim 10^{20}$ m^{-3} and $T \sim 10$ keV. If the plasma is a 50%–50% mixture of D–T, how much fusion power would be produced in JET and what is Q? What is the weight of the plasma? How much does the confinement time have to be scaled up to achieve break-even? How much does the volume have to be scaled up to achieve 2000 MW thermal power output at break-even? In order to achieve this volume scale-up, how much does the linear dimension have to be scaled up?

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